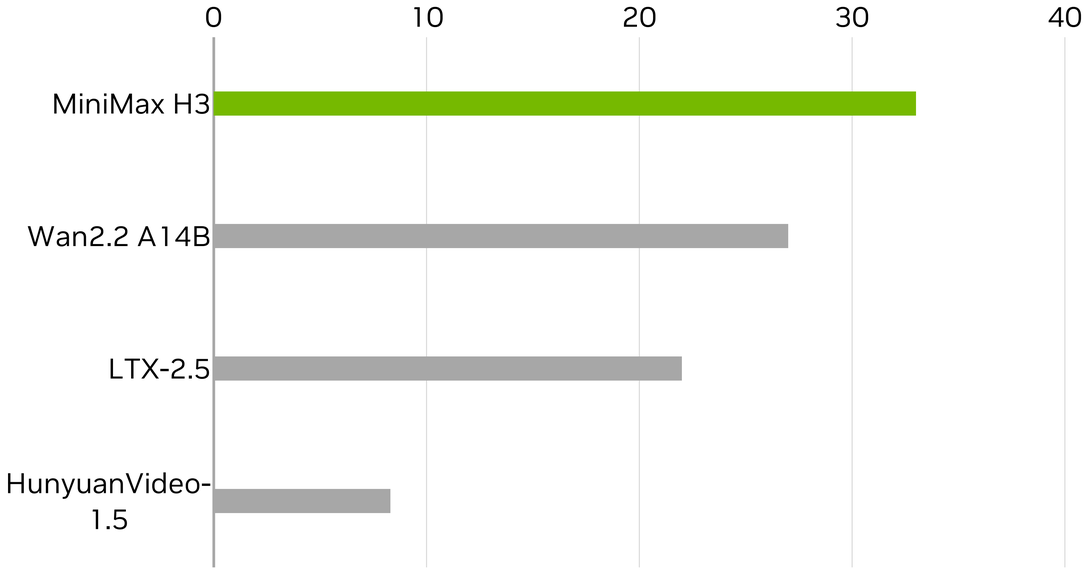
Toward Real-Time Generation With MiniMax H3

8× RTX PRO 5000 · vLLM-Omni

Shunyang Li | NVIDIA Devtech | 1 Nov 2026

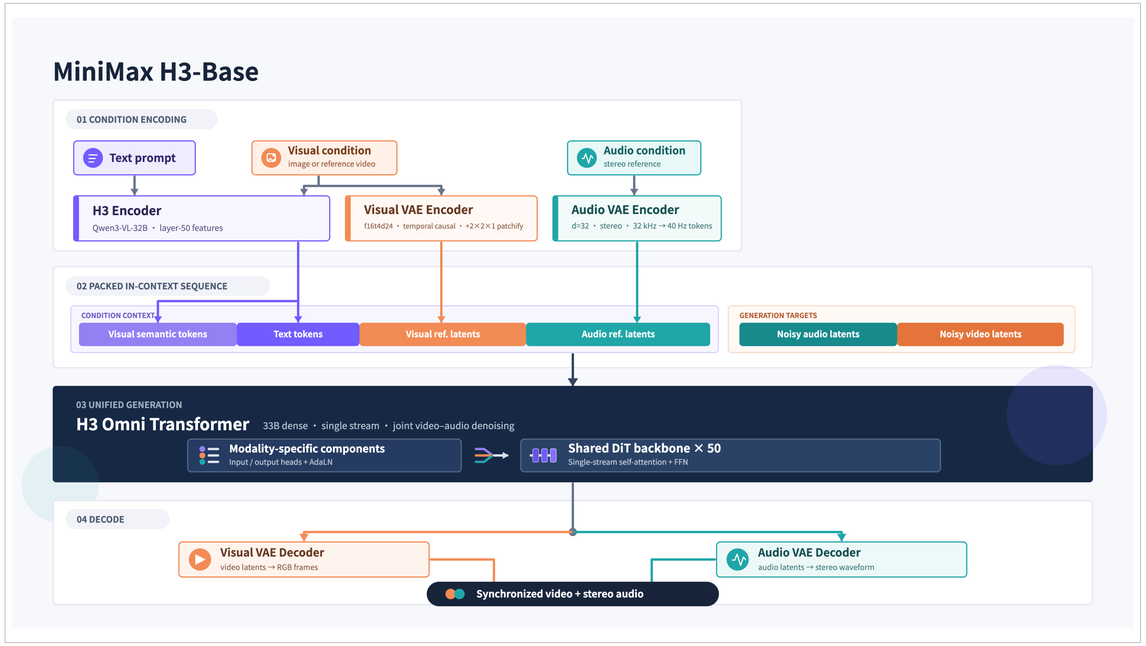
Real-Time Ref2VA Raises the Systems Bar

A 124-frame clip at 24 FPS is real time only when end-to-end latency stays below 5.167 s (RTF < 1)



Generative Backbone · Full Parameters (B)

RTF = latency ÷ media duration; the 124-frame deadline is 5.167 s. H3 has the largest dense backbone in this peer set; Wan2.2 has 27B total and 14B active parameters



Model Capability · Native 768P · 33B Dense

A 32B encoder feeds the 33B Omni Transformer and visual/audio VAEs. Fifty transformer blocks are evaluated at each of 50 baseline steps.

Sources: MiniMax H3, Wan2.2, LTX-2.5 and HunyuanVideo-1.5 official releases · author 960×544 Ref2VA workload

SM120 Brings FP4 Compute; PCIe Limits Scale

All discrete SM120 cards pair fifth-generation Tensor Cores with GDDR7 and PCIe Gen5; none provides GPU-to-GPU NVLink

SKU	Compute	Memory	GPU I/O / Fabric
RTX PRO 5000 ×8 Blackwell	SM120 · 110 SMs 5th-gen Tensor · FP4	72 GB GDDR7 ECC 48 GB variant	PCIe Gen5 ×16 Planned: RDMA via NIC*
RTX PRO 6000 WS / Server	SM120 · 188 SMs 5th-gen Tensor · FP4	96 GB GDDR7 ECC	PCIe Gen5 ×16 No GPU NVLink
GeForce RTX 5090	SM120 · 170 SMs 5th-gen Tensor · FP4	32 GB GDDR7	PCIe Gen5 ×16 No NVLink
DGX Spark · GB10	SM121 · 48 SMs 5th-gen Tensor · FP4	128 GB unified LPDDR5X	NVLink-C2C in SoC 200 Gb/s ConnectX-7

* Planned system path; GPUDirect RDMA requires a compatible NIC and PCIe topology · Sources: NVIDIA product specs, CUDA GPU table, GPUDirect RDMA guide

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Baseline: Latency Breakdown

E2E: 213.402 s · Generated video: 960 × 544, 5.167 s

Reference configuration

- 8 × PRO 5000 GPUs (SM120)
- Original Ref2VA
- 50 steps · BF16 · dense attention
- DiT TP1 / Ulysses8
- Text encoder TP8 · VAE tiles8
- Native VAE · FP16 decode

GPU work by operator

	Mean s / GPU	Share
NCCL · includes waiting	122.182	**57.21%**
Attention	47.105	**22.06%**
GEMM	39.650	**18.57%**
Norm / activation / elementwise	2.288	1.07%
Layout / device copies	1.929	0.90%
CUDA memcpy / memset	0.064	0.03%
Convolution	0.192	0.09%
Other kernels	0.143	0.07%

Nsight GPU work includes overlap and waiting; shares exclude CPU work and idle gaps.

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Three Ideas Behind Faster Generation

Technical overview · detailed algorithms and source excerpts remain available on hidden slides

Less model work

Few-step distillation reduces evaluations.
Sparse attention concentrates work on selected regions.
Quantized inference lowers arithmetic and memory cost.

Overlap across eight GPUs

Ulysses distributes attention across heads.
RDMA carries the exchanges.
Head bands overlap transfer with attention compute.

Keep the output path moving

CUDA Graph replays stable head-band compute.
Fused packing reduces intermediate copies.
NVFP4 VAE and MP4 pre-encoding reduce output cost.

These are optimization families. Use each demo’s own recipe and measurements; the four-step avatar and slide 15 are distinct workloads.

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Optimization Path: E2E Speedup

960 × 544 · 5.167 s output · 8 GPUs · standalone H3

Stage	Cumulative change	Median (s)	E2E speedup vs. previous	RTF
00	Baseline · original BF16 / dense / 50 evaluations	213.402	—	41.304
01	+ PDD · 50 → 8 evaluations	44.327	4.814×†	8.579
02	+ Trained VSA · native 8-step model / runtime	25.915	1.710×	5.016
03	+ QAD NVFP4 DiT / native FP4 attention	19.460	1.332×	3.767
04	+ RDMA / direct layout / V-gate overlap	6.832	2.848×	1.322
05	+ 2 / 3 / 2 head pipeline	6.407	1.066×	1.240
06	+ Compute CUDA graphs	6.193	1.035×	1.199
07	+ 36-layer NVFP4 VAE / MP4 pre-encoding	5.420	1.143×	1.049
08–12	**+ Output and data-movement optimizations**	**4.922**	**1.101×**	**0.953**

08–12: fused tile / AAC preparation, direct I/O packing, compact routing and Q/K/V copy removal.

E2E speedup = previous displayed median ÷ current median. 08–12 compares the final result with stage 07.

Final median: 4.922 s · recheck: 4.927 s · 6/6 measured requests below 5 s

† Recorded median ratio; baseline refreshed separately (n = 2). Stages 01–12 retain n = 3 each; no isolated 50→8-step gain claimed.

RTF = latency / 5.167 s. Standalone H3; ASR / LLM excluded. Model, schedule and precision changes are not all lossless.

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Live Avatar: From a Question to a Video

960 × 544 · 24 FPS · 8 × SM120 GPUs · first window 5.167 s / each continuation adds 4.958 s

EXAMPLE · INTRODUCE YOURSELF

“Hi, could you introduce yourself?”

User question translated from the recorded demo

Microphone
Spoken question

→

ASR
Speech → text
Qwen3-ASR-0.6B

→

LLM
Text → short reply
Qwen3-4B

→

H3

→

Playback
Native audio + video
~5-second windows

LLM reply text

“Hi, I’m Miles. I help with simple, clear answers.”

H3 prompt · English excerpt

The man says exactly this English dialogue
in his reference voice:
"Hi, I'm Miles. I help with simple, clear answers."
... His lips follow the words.

A text reply becomes a speaking avatar clip.

Reference image



Input portrait supplied to H3

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Continuation: Two Clips, One Timeline

The previous ending guides the next clip. Only new content is appended.

CHUNK A / LAST FRAME



CHUNK B / FIRST RETAINED FRAME



CONTINUOUS PLAYBACK

Trim B’s overlapping start

Chunk A · 5.167 s | Chunk B · +4.958 s

Recent audio–video context conditions the next generation; matching audio is trimmed with the overlap.

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Live Avatar: Recorded Conversation

74.55 seconds · user-recorded microphone conversation · native H3 audio + video



Play the recorded interaction · fullscreen avatar, live captions and timing overlays

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Takeaways

01 Multi-GPU optimization starts with topology

Map GPU/NIC links and locate communication bottlenecks.
Use available bandwidth efficiently, or reduce data transfer.
Communication is a key source of lossless speedups.

02 Evaluate quantization case by case

Lossy quantization behaves differently across models and workloads.
Measure quality and latency on representative cases.

03 Next: real-time response and richer interaction

Autoregressive diffusion for lower response latency.
Prompt enhancers for richer interaction.

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